

Edmonton Urban Efficiency

Final Retrospective & Technical Reflection

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December 5, 2025

Introduction: The Vision of Auditable Efficiency

My journey with the "Edmonton Urban Efficiency" project began with a fundamental question about governance and engineering: *How can we ensure that public budget allocation is both efficient and auditable?* As a Computer Engineering student living in Edmonton, I wanted to create a system that didn't just show where buildings are, but mathematically proved where they *should* be. My goal was to move beyond static mapping to a dynamic analysis of construction processes, existing building efficiency, and critical infrastructure gaps. While I initially considered analyzing schools and hospitals, I realized that my custom "Grid Based Spatial Analysis Algorithm" was best suited for emergency response optimization. Thus, the project evolved into a focused tool for analyzing Fire and Police station coverage to maximize public safety per dollar spent.

Achievement of Objectives

Reflecting on my original objectives, I can confidently state that I achieved my primary goal, though the scope had to be adapted to the realities of available data.

The Success (Fire & Rescue): I successfully built a fully operational analysis module for Fire Stations. This was possible because I identified and secured two critical datasets: the precise geolocations of stations and a massive historical dataset of over 80,000 "fire response" rows. Having this "sufficient quantity" of data allowed my algorithm to produce stable, meaningful results. The application now effectively identifies high risk zones by correlating incident density with station proximity.

The Pivot (Police & Security): The Police Station module presented a significant engineering challenge. Unlike the fire data, there is no public API for detailed police incident reports. I discovered that the available tools were limited to UI based reporting via ArcGIS with no direct developer access. However, I did not accept this as a dead end. By inspecting the network requests of the city's existing maps, I managed to uncover hidden data endpoints, allowing me to access 30 days of criminal activity and identify police station locations some of which were not even listed on Google Maps. While I could not build the fully historical analysis I dreamed of due to these permission constraints, I successfully implemented a "Beta" version. This process was motivating; it forced me to "know my city" better than any map could show me. I learned that to achieve the perfect result, I would need to engage with city officials for proper API permissions in the future.

Note on Scope: I made the strategic decision to drop the analysis of hospitals and schools. Early testing showed that the variables affecting their efficiency were incompatible with the distance based logic I developed for emergency services. Dropping them allowed me to focus quality over quantity.

Most Valuable Learning: The "Algorithm" Moment

The most profound learning experience of this project was not a specific syntax or library, but the intellectual breakthrough of designing the core algorithm. For days, I struggled with how to mathematically represent "risk" on a map. When I finally derived the **Minimization & Weighted Scoring Logic**, I felt the distinct satisfaction of solving a complex calculus problem. Building this algorithm taught me that spatial logic is universal. While I applied it to Edmonton Fire Stations, I realized that this logic is not limited to one domain. It is a reusable intellectual asset that can be applied to logistics, supply chain optimization (like Geo ERP systems), or retail expansion. This realization shifted my mindset from "coding a school project" to "engineering a scalable solution."

Collaborative Reflection (Task Delegation)

If I were to execute this project again as part of a diverse team, I would clearly delineate roles based on the "Engine vs. Paint" philosophy.

What I would take ownership of: I would retain absolute ownership of the **Backend, Algorithm Implementation, and Data Engineering**. The process of converting raw JSON data into the specific GeoJSON format required by my algorithm was the most critical step. If the data structure is flawed, the map is meaningless. My passion lies in this "backend logic" constructing the mathematical engine that drives the decision support system.

What I would delegate: I would delegate the **Frontend Design and UI/UX**. While I am capable of building interfaces, I believe that modern AI tools or a specialized frontend developer could handle the visual layer (CSS styling, responsive adjustments) much more efficiently. My time is better spent optimizing the propagation speed of the algorithm than tweaking the color gradients of the sidebar.

Navigating Project Management: From Chaos to Clarity

My project management journey was a transition from "ambiguity" to "simplification." In the beginning, I was overwhelmed. I signed up for every platform, tried dozens of random code examples, and researched every geospatial library available. This lack of focus created a chaotic environment that demotivated me. The turning point came when I applied a simple tactic: **"Simplify the Problem."** I stopped trying to build "everything" and focused entirely on one question: *What data do I actually have?* Once I gathered the Fire Station data, the path became clear. I realized I couldn't force the data to fit my app; I had to shape my app to fit the data. I converted my raw datasets into GeoJSON to align with my algorithm. This experience taught me that in engineering, **clarity comes from constraints**. Defining what I *could* do with the data I *had* saved the project.

Choice of Public Facing Artifact

I chose to deploy a **live, interactive web application** rather than a static report because my project is inherently **Spatial**. You cannot explain the "efficiency of a grid" on a piece of paper. A stakeholder needs to click on a specific square kilometre to see the "Risk Score" and understand the variables at play. Furthermore, I wanted to build a **Sustainable Tool**. A PDF report is "dead data" it represents the state of Edmonton in 2025. By building a web app connected to APIs (even with the current limitations), I created a system that can theoretically work five years from now without changing the code. As long as the data source updates, the tool updates. This architectural decision aligns with my goal of creating a lasting contribution to the open data community.

Impact of the Emerging Talent Program

The **Data Science Lifecycle** taught in the Emerging Talent program was my safety net. In previous projects, I would dive straight into coding, which often led to dead ends. This time, I spent significant energy on the **"Problem Identification"** phase. Forcing myself to define the problem *inefficient budget allocation due to static planning* before writing a single line of code saved me weeks of wasted effort. It kept me from going down the "wrong path" when the technical challenges with the Police API arose.

Conclusion & Future Vision

This capstone project has solidified my passion for the intersection of **Geospatial AI** and **Logistics**. I see a massive potential in applying these "auditable efficiency" algorithms not just to city planning, but to the private sector specifically in **Geo ERP (Enterprise Resource Planning)** systems. I am motivated to continue contributing to open data platforms voluntarily, helping cities unlock the value hidden in their datasets. "Edmonton Urban Efficiency" is just the first iteration of a career I intend to build around optimizing the physical world through digital logic.